REPCURVE / RESEARCH & PRODUCT DESIGN

RepCurve: exercise evidence review and expansion plan

Evidence reviewed through October 11, 2026. Prepared for a fitness app serving adults training at home, in a gym, or across both settings.

Recommended product direction

Keep the home-first foundation. Add scalable pulling, knee-flexion hamstring work, beginner entry options, and a complete set of common gym movements. Organize the engine around movement families, muscle targets, equipment and progression within an exercise; retain muscle groups as a browsing view.

The current list has 44 strength exercises, but several are variations of the same task. Its most consequential omission is knee-flexion hamstring training: all three current hamstring entries are Romanian deadlifts. Other gaps include easy pulling, direct posterior-deltoid options, lateral trunk work, a no-dumbbell calf entry point, and almost the entire machine/cable/barbell ecosystem.

This proposal contains 62 additions in two tiers: 33 for the next release and 29 optional additions. With all tiers implemented, the catalog would contain 106 strength records, including one retired legacy door-row record, leaving 105 selectable records. It also defines eight cardio modalities with steady and interval formats where appropriate. These are choices for the planner, not a prescription to increase exercises per workout.

The strongest evidence supports progressive training, appropriate effort and volume, and consistency. Evidence that one named exercise is universally best is much thinner. Recent trials support specific additions, particularly seated leg curls, leg extensions, and familiar gym alternatives, without establishing a mandatory exercise list for every user.

Five changes to prioritize

1. Add a hamstring curl family alongside the existing RDL family.
2. Make assisted vertical pulls and stable rows available before unassisted pull-ups or renegade rows.
3. Treat machines, cables and free weights as parallel choices selected for the user and task.
4. Replace universal easiest-to-hardest ladders with load progression and optional movement branches.
5. Make the planner respect actual session time, including both sides, warm-ups, setup and equipment transitions.

Review method and interpretation

This is a targeted narrative literature review and product audit, not a preregistered systematic review. It used web-accessible PubMed/PMC records, original journal articles, and ACSM/WHO material. Searches prioritized 2024-2026 syntheses and longitudinal trials, then retained older exercise-specific studies where they directly addressed a catalog decision. No new pooled effect estimate was calculated.

Search themes included resistance-training prescription; volume and frequency; proximity to failure; supersets and low-volume training; machine versus free-weight and elastic resistance; exercise variation; hamstring, quadriceps, calf, glute and shoulder hypertrophy; motion range; and aerobic training. Records published or available online by the review date were eligible. Duplicate preprints and final publications were not treated as independent studies; the final 2023 hip-thrust/squat paper was used.

The evidence register contains 24 selected sources. Access varied: some publisher full texts were available, while others were limited to abstracts or indexed publisher/PubMed records. The register states the access level. Search-result crawl dates were not treated as publication dates. The ACSM position stand is a 2026 publication with a search cutoff in October 2024, so newer studies were examined separately.

How to read the recommendations

- Broad support: repeated evidence across reviews supports a general principle, such as the effectiveness of resistance training or the value of adjustable weekly volume.
- Specific support: a controlled training study supports a particular exercise comparison, within its tested population and protocol.
- Product inference: a practical choice derived from anatomy, equipment, accessibility, time cost and broader evidence, without a direct trial of the exact app implementation.

These labels are not formal GRADE assessments. An exercise can be a valuable inclusion without having an individual randomized trial. Conversely, a single favorable trial does not justify calling it the best exercise for everyone.

Most exercise-comparison studies were short and relatively small, often involving young or untrained adults. Evidence from trained users, older adults, and women is uneven. The catalog recommendations concern general adult fitness; they do not validate injury rehabilitation, pregnancy/postpartum programming, or pediatric training algorithms. Reviews also overlap in their included studies, so their participant counts cannot be added together.

What the evidence does not establish

There is no validated universal ranking from incline push-up to knee push-up to floor push-up, from chin-up to pull-up, or from bilateral to single-leg RDL. Nor does listing a secondary muscle prove that every set provides the same stimulus as direct training of that muscle. Acute electromyography, soreness and feeling a burn should not determine exercise rankings or substitute for longitudinal outcomes.

Findings that should change RepCurve

1. Keep home training fully supported

Home equipment can provide meaningful resistance-training benefits. The push-up comparison supports the existing chest foundation, and elastic resistance has useful strength evidence. The product problem is ensuring sufficient challenge and a viable progression when equipment is limited, particularly for the back and hamstrings. [R01, R09, R10]

For a true no-equipment user, acknowledge that progressively loadable pulling is constrained. Prone arm movements or a dead bug are not equivalent replacements for a row or pulldown. Offer a transparent coverage limitation and the smallest useful equipment path, such as bands plus an appropriate anchor, rather than fabricating balanced pulling coverage.

2. Make gym equipment a first-class choice

Machine and free-weight training can both build muscle. Strength gains are partly specific to the practiced movement, so a user seeking a better barbell bench press needs bench-press practice, while a general fitness user can choose a machine press. [R08]

Add familiar machines and cables before investing in many exotic bodyweight skills. Stable options often make loading and performance tracking simpler; that is a practical product advantage, not a claim that machines are universally safer or biologically superior.

3. Add curls for hamstrings and extensions for quads

An RDL trains the hamstrings through a hip-dominant task; a leg curl supplies a different knee-flexion task. Add sliding curls or walkouts at home and seated/lying curls in the gym. Seated-versus-prone curl research favors the seated condition for several hamstring regions, but does not directly prove sliders outperform RDLs. [R11]

The 2026 knee-extension/leg-press trial found different regional responses. Leg extension therefore deserves a place as complementary quad work, especially for a muscle-growth goal. It need not displace the compound exercise in a severely time-limited session. [R12]

4. Improve shoulder and calf coverage without inventing superiority

Separate anterior, lateral and posterior deltoid tags. Presses and push-ups already cover much anterior-deltoid work; reverse flies and reverse pec-deck options add a direct posterior-deltoid route. The 2025 lateral-raise trial supports both dumbbell and cable versions under matched motion ranges. [R16]

Keep a straight-knee calf raise as the main calf family, and add bodyweight entry points and loadable gym versions. Bent-knee options can be useful, but a 2023 trial found similar soleus growth between standing and seated conditions; do not claim that seated raises uniquely complete soleus training. [R14]

5. Use a modest, adaptable dose

Recent dose-response work supports diminishing returns as weekly training volume increases. A short-workout app should let users start with a manageable dose and increase it according to progress, recovery and available time. It should not reject a session because it cannot reach a hypertrophy-optimization target. [R02]

The 2025 single-set trial supports a low-volume fallback, but involved nine exercises twice weekly; it does not directly validate every abbreviated workout. Offer a short version of the planned workout and preserve

the next session in the sequence. [R06]

6. Teach effort, preserve rest and use supersets selectively

Training closer to failure matters more for hypertrophy than for strength in the exploratory RIR evidence. Two or three repetitions in reserve is a useful starting product rule for many ordinary work sets, not a proven universal optimum. Very light resistance usually needs more repetitions and substantial effort to remain challenging. [R07]

Supersets can shorten training, especially when exercises do not heavily compete. Pair a press with a row, or an upper-body movement with a lower-body movement, when the user can maintain performance. Allow extra rest when breathing or local fatigue interferes. In a crowded gym, two separate machines may be a poor practical pair even when the physiology is sensible. [R04, R18]

7. Make technique stable before making it elaborate

Use repeatable, controlled and tolerable motion ranges. Research on long-muscle-length training is useful but does not establish that lengthened partials should replace all full-range exercise. The 2025 upper-body trial found similar measured adaptations for the two approaches. [R17]

Preserve a small exercise selection across several weeks so performance is interpretable. Switch for pain, missing equipment, stalled fit, preference or a planned goal change. Advanced methods can remain optional; recent review evidence does not show a consistent advantage over conventional training for the general user. [R20, R21]

8. The newest publication is not always the strongest evidence

The August 2026 review in the Journal of Bodywork and Movement Therapies includes only six studies in its qualitative synthesis. Its emphasis on volume and effort aligns with broader evidence, but it should carry less weight than a larger, directly relevant synthesis. Its publication date is newer than its April 2025 search cutoff. [R24]

For the app, track work sets, effort, movement configuration and adherence. Do not optimize total kilograms lifted across unlike exercises: a leg press, split squat and cable curl do not have comparable tonnage-to-growth relationships.

Audit of all 44 current strength exercises

Keep = a normal candidate after user-specific filtering. Optional = a useful branch, not an automatic promotion. Advanced = opt-in with prerequisites. Retire = keep historical records but exclude the generic entry from new automatic plans. The judgments below are product recommendations, not rankings established by comparative trials.

Chest

Exercise	Decision and correction
Incline push-up	Keep: Record hand-support height; knee push-ups are a parallel option, not the next fixed level.
Knee push-up	Keep: Useful regression; adjust repetitions and range before assuming a harder variation is needed.
Push-up	Keep: Main home press; progress reps, leverage or external load when needed.
Feet-elevated push-up	Optional: Optional press variation with more shoulder demand; not a mandatory destination.
Dumbbell floor press	Keep: Useful without a bench; shorter bottom range is a tradeoff, not a reason to exclude it.
Dumbbell bench press	Keep: Keep independent load history from floor press; bench changes the motion range.

Shoulders

Exercise	Decision and correction
Dumbbell lateral raise	Keep: Tag lateral deltoids specifically; this is not interchangeable with overhead pressing.
Pike push-up	Optional: Skill-dependent vertical press; not a universal beginner shoulder exercise.
Feet-elevated pike push-up	Advanced: Advanced variation; retain only as an opt-in progression.
Dumbbell overhead press	Keep: Tag anterior deltoids and triceps; seated and standing setups need distinct metadata.
Single-arm dumbbell overhead press	Optional: Alternative for preference or loading limits; takes longer and is not inherently a higher level.

Triceps

Exercise	Decision and correction
Dumbbell overhead triceps extension	Keep: Keep; direct overhead-versus-neutral evidence used cables, so dumbbell transfer is an inference.

Back

Exercise	Decision and correction
Band row	Keep: Specify a secure anchor or a separately taught seated foot-braced setup.
Towel door row	Retire: Retire the generic door-based entry from automatic plans; replace with a rated suspension or band setup.
One-arm dumbbell row	Keep: Specify self-braced versus externally supported technique; add support equipment when used.
Dumbbell bent-over row	Keep: A lower-back endurance limit can precede the back-muscle target; offer chest support.
Renegade row	Optional: Classify as row plus anti-rotation; do not automatically replace a stable back-focused row.
Negative pull-up	Advanced: Needs a controlled way to reach the top; unfamiliar eccentric work deserves gradual exposure.
Chin-up	Keep: Parallel grip variation; not a universally easier rung below pull-ups.
Pull-up	Keep: Add assisted options; track body mass and assistance separately from external load.

Biceps

Exercise	Decision and correction
Dumbbell curl	Keep: Keep as a simple direct elbow-flexor exercise; label forearm position.

Quads

Exercise	Decision and correction
Bodyweight squat	Keep: Can become too easy for strength or growth; add external load when available.
Split squat	Keep: Parallel to reverse lunges; support can remove balance as the limiting factor.
Reverse lunge	Keep: Useful option; do not count hamstrings as equal direct work to quads and glutes.
Bulgarian split squat	Optional: Higher balance and setup demands; not obligatory after a split squat.
Goblet squat	Keep: Efficient home compound; arm position or available load can eventually limit progression.
Dumbbell front squat	Keep: Loadable alternative; not a required step after goblet squats.
Dumbbell reverse lunge	Keep: Log repetitions per side and budget time for both sides.
Dumbbell Bulgarian split squat	Optional: Useful with limited dumbbell weight; make stability, tolerance and time explicit.

Glutes

Exercise	Decision and correction
Dumbbell glute bridge	Keep: Progress load on the same pattern; hamstrings are secondary, not a curl substitute.
Dumbbell hip thrust	Keep: Require a stable padded support; a generic chair or step is not always suitable.
Glute bridge	Keep: Beginner hip-extension option; increase challenge when repeated easy reps cease to be productive.
Single-leg glute bridge	Optional: Alternative when load is limited, with a per-side time cost.
Single-leg hip thrust	Optional: Optional branch; do not force it instead of a loadable bilateral thrust.

Hamstrings

Exercise	Decision and correction
Dumbbell Romanian deadlift	Keep: Keep as the main home hinge and add a distinct knee-flexion exercise family.
Staggered-stance dumbbell RDL	Optional: Useful asymmetric-loading branch; record front-leg emphasis and repeat on the other side.
Single-leg dumbbell RDL	Optional: Balance may limit loading; offer external hand support as a tracked variation.

Calves

Exercise	Decision and correction
Dumbbell calf raise	Keep: Define knee position, bilateral versus unilateral technique, and optional step use.

Core

Exercise	Decision and correction
Dead bug	Keep: Progress limb leverage and controlled range; no need to advance to hanging work.
Plank	Keep: Anti-extension; shoulders stabilize. Progress leverage rather than only extending hold duration.
Hollow hold	Optional: Parallel anti-extension branch; not a universal next level after plank.
Hanging knee raise	Optional: Hip flexors are prime movers too; add posterior pelvic tilt when abdominal flexion is intended.
Kneeling ab-roller rollout	Optional: Progress rollout distance gradually; kneeling pads are optional comfort equipment.
Standing ab-roller rollout	Advanced: Advanced opt-in skill; not the default endpoint of core training.

Proposed additions by muscle group

P1 means add in the next release; P2 means optional expansion. The full catalog remains filtered by equipment, skill, goal, tolerance and time. Equipment listed here includes meaningful supports and surfaces. A machine or cable record is a parallel exercise choice, not an upgrade the user must eventually reach.

Many additions are supported by movement coverage and practical usability rather than a direct trial of the exact variation. The JSON records distinguish these inferences from specific tested exercise comparisons.

Chest

Exercise / tier	Equipment and reason
Wall push-up (P1)	wall + standing space. Accessible entry point before elevated-support push-ups.
Incline dumbbell bench press (P1)	dumbbells + adjustable bench. Adds an incline press option; record bench angle.
Barbell bench press (P1)	barbell + plates + bench press rack + flat bench + bench safeties or spotter. Standard gym press with scalable loading and a distinct skill history.
Machine chest press (P1)	chest press machine. Stable, quickly adjustable gym pressing option.
Cable chest fly (P2)	dual cable station + handles. Optional chest isolation when pressing alone does not match the goal.

Shoulders

Exercise / tier	Equipment and reason
Dumbbell reverse fly (P1)	dumbbells. Adds direct posterior-deltoid work; specify torso or bench support.
Band overhead press (P2)	resistance band. Standing-on-band option for users without dumbbells; track band and stance.
Machine shoulder press (P1)	shoulder press machine. Stable vertical press option with modest setup complexity.
Cable lateral raise (P2)	cable station + handle. Preference and resistance-profile alternative to dumbbell lateral raises.
Reverse pec deck (P2)	reverse fly machine. Supported posterior-deltoid and upper-back isolation.
Barbell overhead press (P2)	barbell + plates + rack. Gym strength option; not required for complete shoulder coverage.

Triceps

Exercise / tier	Equipment and reason
Band triceps pressdown (P2)	resistance band + rated high anchor. Simple home triceps option without overhead arm positioning.
Cable triceps pressdown (P1)	cable station + pressdown attachment. Quick gym isolation and alternative to an overhead extension.
Cable overhead triceps extension (P1)	cable station + extension attachment. Directly represented in overhead-position research; adjust setup for comfort.

Back

Exercise / tier	Equipment and reason
Band lat pulldown (P1)	resistance band + rated high anchor. Accessible home vertical pulling before bodyweight pull-ups.
Suspension row (P1)	suspension trainer + rated suspension anchor. Body angle scales difficulty; record foot position and handle height.
Band-assisted pull-up (P1)	pull up bar + assistance band + stable access platform. Bridges the large gap to unassisted pull-ups; assistance varies through the rep.
Chest-supported dumbbell row (P1)	dumbbells + adjustable bench. Reduces the unsupported hinge requirement; useful at home or in a gym.
Lat pulldown (P1)	lat pulldown machine + pulldown attachment. Scalable vertical pull for a wide range of strength levels.
Seated cable row (P1)	seated row station + row attachment. Scalable horizontal pull with straightforward load adjustment.
Chest-supported machine row (P2)	supported row machine. Stable alternative; machine geometry and grip should be recorded.
Assisted pull-up machine (P1)	assisted pull up machine. Adjustable assistance for users not ready for full-bodyweight pulling.

Biceps

Exercise / tier	Equipment and reason
Band curl (P2)	resistance band. Fills the band-only elbow-flexor option.
Dumbbell hammer curl (P2)	dumbbells. Neutral-grip elbow-flexor option, including brachialis and brachioradialis.
Machine preacher curl (P2)	preacher curl machine. Supported, loadable gym curl; arm position and machine fit matter.
EZ-bar curl (P2)	ez bar + plates. Common gym curl alternative; no universal advantage over dumbbells.

Quads

Exercise / tier	Equipment and reason
Sit-to-stand (P1)	stable chair. Accessible squat entry point with a standardized seat height.
Supported split squat (P1)	stable hand support + floor space. Removes some balance demand; can later accept load in the free hand.
Step-up (P1)	stable step + balance support. Functional step pattern; record height and hand support.
Dumbbell step-up (P2)	dumbbells + stable step. Loaded step option with explicit step height and per-side time.
Barbell back squat (P1)	barbell + plates + squat rack + safeties. Standard gym squat; optional for users who prefer machines.
Leg press (P1)	leg press machine. Stable multijoint lower-body option; record machine and seat settings.
Hack squat machine (P2)	hack squat machine. Another loadable squat pattern; not a necessary separate weekly slot.
Smith machine squat (P2)	smith machine. Guided-bar squat alternative with its own load and technique history.
Leg extension (P1)	leg extension machine. Direct knee extension complements compound quad training.
Barbell front squat (P2)	barbell + plates + squat rack + safeties. Goal- and preference-driven squat variation with higher rack-position demands.

Glutes

Exercise / tier	Equipment and reason
Bodyweight hip thrust (P2)	stable shoulder support. Missing intermediate setup for teaching the hip-thrust pattern.
Barbell hip thrust (P1)	barbell + plates + stable shoulder support + bar pad. Scalable gym glute option; include loading and setup time.
Machine hip thrust (P2)	hip thrust machine. Efficient setup alternative when the gym has the machine.
Trap-bar deadlift (P2)	trap bar + plates + clear lifting area. Multijoint strength option involving quads, glutes and trunk; handle height changes the task.
Barbell deadlift (P2)	barbell + plates + clear lifting area. Include for conventional-deadlift goals; not a required hypertrophy or beginner exercise.

Hamstrings

Exercise / tier	Equipment and reason
Band Romanian deadlift (P1)	resistance band. Standing-on-band hinge for users with bands but no dumbbells.
Hamstring walkout (P1)	floor space. No-special-equipment bridge-to-knee-flexion option; begin with short excursions.
Sliding leg curl (P1)	sliders + compatible floor OR towels + smooth floor. Home knee-flexion exercise; range and eccentric-only regressions can scale difficulty.
Barbell Romanian deadlift (P1)	barbell + plates. Loadable gym hinge; rack is optional but affects setup.
Seated leg curl (P1)	seated leg curl machine. High-priority knee-flexion addition with direct hypertrophy evidence.
Lying leg curl (P2)	lying leg curl machine. Useful gym alternative when seated curls are unavailable or unsuitable.

Calves

Exercise / tier	Equipment and reason
Bodyweight standing calf raise (P1)	balance support. Accessible calf entry point; optional step is a separate range modifier.
Single-leg bodyweight calf raise (P2)	balance support. Increases relative load without equipment; balance support remains useful.
Standing machine calf raise (P1)	standing calf machine. Loadable straight-knee calf option with direct pattern evidence.
Seated machine calf raise (P2)	seated calf machine. Bent-knee alternative; do not market it as necessary for soleus growth.
Seated dumbbell calf raise (P2)	dumbbell + stable chair. Home bent-knee option with load placed securely above the knee.

Core

Exercise / tier	Equipment and reason
Side plank (P1)	floor space. Adds lateral trunk endurance; bent-knee and full-leg versions share a family.
Reverse crunch (P1)	floor space. Accessible dynamic abdominal exercise without grip or hanging requirements.
Pallof press (P1)	resistance band + rated chest height anchor OR cable station + handle. Anti-rotation branch; band and cable setups require separate performance contexts.
Suitcase carry (P2)	dumbbell + clear walkway. Grip plus lateral trunk bracing; log load, duration or distance, and side.
Cable crunch (P2)	cable station + rope attachment. Loadable trunk-flexion option for gym users.

Hip accessories and spinal erectors

Exercise / tier	Equipment and reason
Side-lying hip abduction (P1)	floor space. Simple direct lateral-hip option; an accessory, not mandatory for every short session.
Band hip abduction (P2)	loop band + balance support. Standing lateral-hip option; standardize band placement and range.
Hip abduction machine (P2)	hip abduction machine. Scalable gym accessory when lateral-hip work is a goal.
Hip adduction machine (P2)	hip adduction machine. Optional direct inner-thigh work; compounds already provide some coverage.
45-degree back extension (P2)	back extension bench. Teach a defined hip/back-extension technique; targets vary with execution.

Programming for limited time

These are proposed implementation defaults informed by the review, not prescriptions tested as one complete RepCurve program. Adapt to the user's experience, goal, preferences and response.

Starter defaults

Decision	Proposed default
Schedule	Two or three full-body sessions per week for users with limited availability. A split is also valid when it better fits the schedule; matched-volume evidence does not show a clear winner. [R19]
Session menu	Usually four to six exercises. Cover major movement families across the week; do not demand every movement pattern in each short session.
Work sets	Start with one or two work sets per exercise for new or time-limited users. Add sets selectively when time, performance and recovery support it.
Rep range	A practical starting band is 6-15 reps for many compounds and 10-20 for many accessories. Bodyweight/band work may use roughly 10-30 when needed. These are defaults, not effectiveness boundaries. [R03]
Effort	Begin with about 2-3 good repetitions in reserve on typical work sets, using a larger margin while learning. Calibrate the estimate over time; do not make failure mandatory. [R06, R07]
Weekly dose	A starting design target of roughly 4-8 challenging direct-equivalent sets per major muscle is reasonable for this time-limited product, then adjust. Around 10 or more weekly sets may be useful for a muscle-growth emphasis when feasible; no sharp minimum guarantees growth. [R01, R02]
Rest	Begin around 90-180 seconds for demanding compounds and 60-120 seconds for many accessories; allow longer for heavy strength work or performance recovery. Alternating exercises can provide local rest without idle time. [R04, R18]
Warm-up	Budget a few minutes plus easier practice sets when the task/load warrants them. Do not require rope jumping or a lengthy generic routine before every short workout. [R05]
Exercise continuity	Keep useful exercises stable over a several-week block. Review changes deliberately; do not randomize the menu every session. [R20]
Progression	When all prescribed sets reach the top of the rep band with the intended effort and technique on two exposures, try the next available small load increase. If the increment is too large, continue reps or use another controllable progression variable.

The proposed two-exposure rule is an engineering choice intended to reduce overreaction to one unusually good session. The research does not establish it as the uniquely correct progression rule.

Two-session home example

Dumbbells and floor space are sufficient for the main work. A sliding curl also requires a compatible floor and sliders or towels; use walkouts when that setup is absent. Complete each pair for two rounds, with rest adjusted to preserve form and repetitions.

Session A	Session B
Pair 1: goblet squat + dumbbell floor press	Pair 1: dumbbell reverse lunge + dumbbell overhead press
Pair 2: dumbbell RDL + one-arm dumbbell row	Pair 2: sliding leg curl or hamstring walkout + one-arm dumbbell row
If time remains: lateral raise or standing calf raise	If time remains: push-up or side plank, according to the weekly coverage gap

This is a compact general-fitness starting template, not a maximal-growth routine. If using one-arm rows or lunges, both sides count toward the time budget. Avoid pairing demanding hinges and rows if grip or lower-back fatigue makes the second movement unproductive. A supported row helps when that option is available. A 20-30-minute estimate is plausible with efficient transitions, but the app should use measured completion times rather than promise an exact duration.

Two-session gym example

Session A	Session B
Leg press; machine chest press; seated cable row; seated leg curl	Barbell RDL or another suitable hinge; incline dumbbell press; lat pulldown; leg extension
Optional: lateral raise or calf raise	Optional: shoulder press, calf raise or trunk work according to the weekly gap

Begin with two work sets for each main movement. Use straight sets if equipment is occupied; only superset stations that can be used without blocking other users. A user whose goal is barbell squat or bench strength should practice those lifts instead of expecting machine strength to transfer completely. [R08]

A genuinely short fallback

When only 10-15 minutes are available, choose three priority movements from the planned session, perform one or two useful work sets each, and continue the normal sequence next time. Preserve a lower-body movement, a push and a pull when equipment permits; rotate the exact coverage according to recent training. Do not convert an incomplete session into a failed day or automatically compensate with excessive volume later. This fallback is a practical extrapolation from low-volume evidence, not a directly tested recipe. [R05, R06]

Progression, substitutions and data model

Keep progression local to a movement

Family	Useful progression and branches
Horizontal push	Wall or high-incline push-up -> gradually lower hand support -> floor push-up. Knee push-up is an alternative branch. Dumbbell and machine presses form parallel load-based tracks.
Vertical pull	Band pulldown or machine pulldown, plus assisted pull-up practice if wanted. Reduce assistance gradually. Chin-up and pull-up remain separate grip variants; negatives are an optional skill tool.
Squat / lunge	Sit-to-stand -> free squat -> loadable squat. Supported split squat, reverse lunge, step-up and Bulgarian split squat are optional branches; no universal ranking is implied.
Hinge	Learn and load a bilateral hinge. Staggered or supported single-leg versions can increase relative challenge when weight is limited. A less stable single-leg version is not automatically a better hamstring stimulus.
Knee flexion	Short-range walkout or assisted sliding curl -> longer controlled sliding curl. Seated and lying machines have separate load-based tracks.
Core	Maintain distinct anti-extension, lateral-bracing, anti-rotation and trunk-flexion families. Do not place dead bug, hanging knee raise and standing rollout on one compulsory ladder.

Substitution should preserve the intended job

First filter for actual equipment and setup, current restrictions, user-reported tolerance, and skill prerequisites. Then choose a replacement that preserves the intended movement and muscle target as closely as practical. Finally minimize setup, transition and per-side time costs.

A row is usually a closer substitute for another row than for a pulldown. A curl is not a substitute for an RDL solely because both are labeled hamstrings. A renegade row can be useful core work without being the best replacement for a chest-supported back exercise. When no close substitute exists, state the coverage tradeoff in the plan and rebalance later.

Correct the muscle accounting

Keep one primary browsing category if it helps the UI, but store prime movers, secondary movers and stabilizers separately. Direct quadriceps/glute work from squats does not make the squat a full hamstring-training set. A row does not count equally for every shoulder region. Hanging knee raises involve hip flexors as well as abdominal control; pulling-muscle participation is largely supporting the hang.

If the engine uses fractional set credits, make them configurable, versioned approximations rather than biological facts. A starting model might credit a direct target fully and a substantial secondary mover partially; it should not credit every stabilizer the same way. The 2026 dose-response paper supports considering indirect work but does not validate exact coefficients for this catalog. [R02]

Equipment is more than a yes/no label

Store dumbbell counts, maximum load and increments; flat versus adjustable bench; band type and anchor height; rack and safeties; machine type; floor compatibility; stable hand/foot/shoulder supports; and space or noise limits. An incline push-up is bodyweight training with a support requirement, not literally equipment-free. A band row needs a defined anchoring method. Do not recommend a random door, table or chair as a load-rated exercise station.

Home and gym equipment should be separate location profiles, with a session-specific available set. A cable station being busy should trigger a same-session alternative without deleting it from the user's permanent equipment list.

Preserve honest performance histories

Key performance to exercise plus configuration: machine identity, bench angle, handle, support height, assistance band, stance and range. Do not transfer a 40-pound dumbbell load to a 40-pound cable setting as if the tasks are identical. For assisted pull-ups, less assistance is progression; for unilateral exercises, repetitions and loads must be logged consistently per side. Estimate one-repetition maximum only where appropriate, not for arbitrary timed holds or mismatched band setups.

The supplied JSON defines 106 proposed and legacy strength records, equipment logic, movement families, targets, selection notes and source mappings. It is a design seed, not a ready-to-deploy recommendation engine. Instructional media, configuration-specific variants, calibrated time estimates and user-response logic still need implementation and verification.

Cardio, warm-ups and the pain response

Expand modes and separate mode from workout format

Your current list mixes equipment/mode (jump rope), terrain (incline), and session structure (intervals). Store these separately. Walking can be steady or interval-based, and a stationary bike can serve either format. Effort-based prescriptions are sensible, but the scale must be defined consistently and reinforced with the talk test.

Mode	Role and setup
Walking	Includes flat and incline walking; incline is a setting, not a separate required mode.
Jogging or running	Walk-run intervals provide an entry route; do not equate slow speed with low individual effort.
Cycling	High-priority home/gym cardio addition.
Elliptical	Useful alternative when running is not preferred.
Rowing ergometer	Needs technique familiarization and contributes lower-back, grip and posterior-chain fatigue.
Stair climber or stair walking	Budget leg fatigue; stairs require balance and environment checks.
Jump rope	Skill and repeated impact matter; possessing a rope does not make it the default warm-up.
Indoor marching or low-impact movement	Accessible weather-independent option; verify that effort reaches the intended intensity.

For general aerobic health, show progress toward the WHO weekly activity targets while recognizing that smaller amounts count. Include everyday walking instead of requiring all activity to occur inside an app workout. Strength and aerobic minutes should be tracked separately. [R22]

A practical cardio interface can use easy, moderate and hard effort descriptors: easy allows relaxed conversation; moderate permits talking but makes singing difficult; vigorous work permits only short phrases. These talk-test distinctions follow CDC intensity guidance. Any numerical 0-10 labels should be tied to a named scale rather than mixed with strength RIR. A very short strength session is not evidence that the aerobic recommendation has been met.

Recent real-world HIIT evidence supports offering intervals, but does not make HIIT necessary or establish a best modality for everyone. Cycling is a particularly useful addition for users who prefer to avoid repeated running/jumping impacts. Start new users with tolerable steady work or gentle intervals; progression depends on response. Rope jumping remains optional, including in warm-ups. [R23]

Make “It hurt” a session-control action

The existing promise that the app swaps movements which clash with an injury is too strong if it relies only on a muscle label or a generic injury blacklist. Pain can depend on load, range, speed, fatigue, setup and the person. A replacement that targets the same muscle is not automatically pain-free or appropriate.

Proposed behavior: stop the current set; retain the pain report with body location and movement configuration; prevent automatic re-exposure in that session; offer an optional lower-demand or different-pattern alternative only after a tolerance check, or allow skipping the slot. Do not automatically add weight or advance a variation after a pain flag. New severe symptoms or concerning associated symptoms should end the exercise flow rather than trigger another exercise suggestion.

Avoid diagnosing the injury or promising that a replacement is safe. Keep “too hard,” “could not balance,” “equipment unavailable,” “disliked it,” and “it hurt” as separate feedback events because they require different responses. These are product safeguards, not a validated treatment algorithm.

Release sequence and acceptance checks

First: repair selection logic and metadata

Correct supports and anchors, retire the generic towel-door row from new plans, and replace universal ladders with movement families. Preserve old logs. Add location-specific equipment and per-side timing.

Next: ship the 33 P1 additions

Prioritize accessible entry points, hamstring curls, assisted pulling, gym compounds and useful isolation. Keep each user's menu small, with shorter and ordinary versions of the same plan.

Then: ship the 29 P2 additions as needed

Use these for preference, equipment compatibility and specialized goals. Extra machine variants, advanced skills, carries and hip accessories remain optional.

Acceptance checks for the planner

- Equipment-free plans have accessible entry points and represent pulling limitations honestly.
- Band exercises match the available anchors.
- Gym beginners can receive a coherent machine-based plan.
- Dumbbell users can progress without compulsory balance skills.
- Plans distinguish hinges from curls and shoulder regions from one another.
- Time estimates include both sides, setup and transitions.
- Pain reports stop automatic progression and re-exposure.
- Strength goals receive specific lift practice; general fitness retains flexibility.
- Claims distinguish direct evidence from practical extrapolation.

The central recommendation is to expand the choices available to the planner while keeping each user's workout small, repeatable and progressively challenging.

Evidence register and reading list

The entries below supply the source links, study design, useful finding and practical limit behind the recommendations. Broad programming claims draw on multiple overlapping syntheses; these are not 24 independent datasets.

[R01] Currier et al. / ACSM (2026)

American College of Sports Medicine Position Stand. Resistance Training Prescription for Muscle Function, Hypertrophy, and Physical Performance in Healthy Adults: An Overview of Reviews. Medicine & Science in Sports & Exercise 58(4):851-872.

Design: Overview of 137 reviews; >30,000 participants; search through October 2024.

Finding: Progressive resistance training improves multiple outcomes. Regular training of major muscle groups is the foundation; prescriptions should reflect the goal.

Application and limit: Broad support for home and gym modes; strongest evidence for programming principles, not a ranking of individual exercises. The 2026 publication does not itself include all 2025-26 research.

Access: Indexed abstract plus official ACSM summary. Source | DOI

[R02] Pelland et al. (2026)

The Resistance Training Dose Response: Meta-Regressions Exploring the Effects of Weekly Volume and Frequency on Muscle Hypertrophy and Strength Gains. Sports Medicine 56:481-505; online December 4, 2025.

Design: 67 studies, 2,058 participants; about 79% male; mean age about 25.

Finding: More weekly sets predicted greater gains with diminishing returns. Frequency had a clearer independent relationship with strength than with hypertrophy. Fractional accounting for indirect sets fitted the data best.

Application and limit: Support adjustable weekly volume and distinguish direct from indirect work. A 0.5 contribution is a modeling convention, not a universal biological conversion.

Access: Publisher abstract. Source | DOI

[R03] Currier et al. (2023)

Resistance training prescription for muscle strength and hypertrophy in healthy adults: a systematic review and Bayesian network meta-analysis. British Journal of Sports Medicine 57:1211-1220.

Design: Strength network: 178 studies / 5,097 participants; hypertrophy: 119 / 3,364.

Finding: Many prescriptions improved outcomes. Heavier loading was especially relevant to maximal strength; multiple sets were useful for hypertrophy.

Application and limit: Provide strength and muscle-growth settings without making a single rep range the only effective choice. Network rankings do not prove one universal best program.

Access: Indexed abstract and publisher record. Source | DOI

[R04] Zhang et al. (2025)

Superset Versus Traditional Resistance Training Prescriptions: A Systematic Review and Meta-analysis Exploring Acute and Chronic Effects on Mechanical, Metabolic, and Perceptual Variables. Sports Medicine 55:953-975.

Design: 19 studies, 313 participants; mixed acute and longer-term outcomes.

Finding: Supersets shortened sessions and generally preserved adaptations, while raising perceived effort. Same-muscle pairings could reduce completed training volume.

Application and limit: Offer noncompeting pairs as a time-saving option. The chronic evidence is smaller than the total study count; preserve performance and recovery.

Access: PubMed abstract. Source | DOI

[R05] Iversen et al. (2021)

No Time to Lift? Designing Time-Efficient Training Programs for Strength and Hypertrophy: A Narrative Review. Sports Medicine 51:2079-2095.

Design: Narrative synthesis focused on limited-time training.

Finding: Bilateral multijoint exercises, selective warm-ups, and economical set structures can reduce the time burden.

Application and limit: A useful design framework, not a trial validating a particular 20-minute app routine. Avoid assuming unilateral work is always the best upgrade.

Access: PubMed abstract and indexed full-text sections. Source | DOI

[R06] Without Fail trial (2025)

Without Fail: Muscular Adaptations in Single-Set Resistance Training Performed to Failure or with Repetitions-in-Reserve. Medicine & Science in Sports & Exercise 57(9):2021-2031.

Design: 42 trained young men and women; one set of nine exercises, twice weekly, for eight weeks.

Finding: Both failure and two-repetitions-in-reserve groups improved most outcomes. Some muscle-growth outcomes modestly favored failure; strength changes were similar.

Application and limit: Justifies a low-volume fallback. It does not establish that one set maximizes gains or that a shorter four-exercise plan reproduces this trial.

Access: PubMed abstract. Source | DOI

[R07] Robinson et al. (2024)

Exploring the Dose-Response Relationship Between Estimated Resistance Training Proximity to Failure, Strength Gain, and Muscle Hypertrophy: A Series of Meta-Regressions. Sports Medicine 54:2209-2231.

Design: Exploratory meta-regressions; repetitions in reserve were estimated from study descriptions.

Finding: Training closer to failure was associated with more hypertrophy; strength gains were less sensitive across the studied range.

Application and limit: Teach effort and use challenging sets, but do not force failure or describe a precise RIR cutoff as settled science.

Access: PubMed abstract. Source | DOI

[R08] Free-weight versus machine meta-analysis (2023)

Effect of free-weight vs. machine-based strength training on maximal strength, hypertrophy and jump performance - a systematic review and meta-analysis. BMC Sports Science, Medicine and Rehabilitation.

Design: 13 studies; only five contributed hypertrophy outcomes.

Finding: No clear hypertrophy advantage for either mode; strength improvements were substantially specific to the trained and tested mode.

Application and limit: Machines belong in the main gym pool. This is not proof that every machine and free-weight variation is equivalent, or that one mode is universally safer.

Access: Publisher abstract/full-text sections. Source | DOI

[R09] Lopes et al. (2019)

Effects of training with elastic resistance versus conventional resistance on muscular strength: A systematic review and meta-analysis. SAGE Open Medicine.

Design: Systematic review and meta-analysis of elastic versus conventional resistance.

Finding: Elastic resistance can improve strength with broadly comparable results to conventional resistance in the included comparisons.

Application and limit: Bands are a legitimate home option. Evidence here concerns strength, not proof of equal hypertrophy for every band exercise. Standardize anchor position and band setup.

Access: Indexed abstract. Source | DOI

[R10] Kikuchi and Nakazato (2017)

Low-load bench press and push-up induce similar muscle hypertrophy and strength gain. Journal of Exercise Science & Fitness.

Design: 18 young men; eight weeks, twice weekly; load-matched push-ups and low-load bench press.

Finding: The groups had similar muscle-growth and strength improvements under the tested conditions.

Application and limit: Keep scalable push-ups as a principal home chest exercise. Small, narrow trial; do not extrapolate to equal maximal-strength outcomes at every ability level.

Access: Indexed abstract. Source | DOI

[R11] Maeo et al. (2021)

Greater Hamstrings Muscle Hypertrophy but Similar Damage Protection after Training at Long versus Short Muscle Lengths. Medicine & Science in Sports & Exercise 53(4):825-837.

Design: Within-person comparison of seated versus prone knee-flexion training.

Finding: Whole-hamstring volume increased more with seated than prone curls (about 14% versus 9%); the difference involved the biarticular muscles.

Application and limit: Add knee-flexion work and favor seated curls when available and comfortable. This was not a direct comparison of curls versus RDLs or sliders.

Access: Indexed abstract. Source | DOI

[R12] Kinoshita et al. (2026)

Hypertrophic Effects of Single- versus Multi-Joint Exercise: A Direct Comparison between Knee Extension and Leg Press. Medicine & Science in Sports & Exercise 58(7):1566-1580; online February 2026.

Design: 17 untrained adults; opposite legs trained for 12 weeks; MRI outcomes.

Finding: Rectus femoris growth favored knee extension (13.2% versus 1.1%); other quadriceps regions and total quadriceps were more similar. Leg press also grew gluteus maximus and adductor magnus.

Application and limit: Include leg press as a compound option and leg extension as complementary quad work. The small single study does not make isolation mandatory in every short workout.

Access: Indexed PubMed abstract and publisher record. Source | DOI

[R13] Maeo et al. (2023)

Triceps brachii hypertrophy is substantially greater after elbow extension training performed in the overhead versus neutral arm position. European Journal of Sport Science 23(7):1240-1250; online 2022.

Design: 21 adults; opposite arms; cable extensions twice weekly for 12 weeks.

Finding: Overhead extensions produced greater triceps growth than neutral-arm extensions in this protocol, particularly in the long head.

Application and limit: Retain the overhead pattern and add a cable version plus pressdowns as an alternative. The direct trial used cables, not the existing dumbbell variation.

Access: PubMed abstract. Source | DOI

[R14] Kinoshita et al. (2023)

Triceps surae muscle hypertrophy is greater after standing versus seated calf-raise training. Frontiers in Physiology 14:1272106.

Design: 14 untrained adults; opposite legs; 12 weeks.

Finding: Standing raises produced substantially more gastrocnemius growth; soleus growth was similar between conditions.

Application and limit: Prioritize a straight-knee calf option. Seated raises remain useful alternatives, but the trial does not show that seated work is required for soleus growth.

Access: Publisher full text. Source | DOI

[R15] Plotkin et al. (2023)

Hip thrust and back squat training elicit similar gluteus muscle hypertrophy and transfer similarly to the deadlift. Frontiers in Physiology 14:1279170.

Design: Nine-week randomized training comparison using MRI; final peer-reviewed version.

Finding: Gluteal hypertrophy was similar, thigh growth favored squats, and strength gains favored the practiced exercise.

Application and limit: Squats and hip thrusts are legitimate choices for different priorities. Acute muscle-activation rankings alone should not determine the catalog.

Access: Publisher abstract/full-text sections. Source | DOI

[R16] Larsen et al. (2025)

Dumbbell versus cable lateral raises for lateral deltoid hypertrophy: an experimental study. Frontiers in Physiology 16:1611468.

Design: 24 completers; trained men and women; opposite arms; eight weeks; matched motion range.

Finding: Dumbbells and cables produced similar lateral-deltoid growth under these conditions.

Application and limit: Add cables for preference and availability, not as a superior level unlocked after dumbbells. Different motion ranges were not tested.

Access: Publisher full text. Source | DOI

[R17] Wolf et al. (2025)

Lengthened partial repetitions elicit similar muscular adaptations as full range of motion repetitions during resistance training in trained individuals. PeerJ.

Design: 30 trained participants; opposite arms; eight-week upper-body intervention.

Finding: Lengthened partials and full-range training produced similar measured arm hypertrophy and pulldown strength-endurance changes.

Application and limit: Use controlled, tolerable full range by default; retain lengthened partials as an optional technique. This is not evidence for replacing all full-range training.

Access: PubMed abstract. Source

[R18] Singer et al. (2024)

Give it a rest: a systematic review with Bayesian meta-analysis on the effect of inter-set rest interval duration on muscle hypertrophy. Frontiers in Sports and Active Living 6:1429789.

Design: Nine studies; wide and overlapping estimates.

Finding: Results suggested a small benefit from resting more than 60 seconds; additional differences above 90 seconds were not clearly detected.

Application and limit: Do not achieve short sessions by indiscriminately removing rest. Adjust rest to restore useful performance rather than treating one duration as a universal optimum.

Access: Publisher full text. Source | DOI

[R19] Ramos-Campo et al. (2024)

Efficacy of Split Versus Full-Body Resistance Training on Strength and Muscle Growth: A Systematic Review With Meta-Analysis. Journal of Strength and Conditioning Research 38(7):1330-1340.

Design: 14 studies, 392 participants.

Finding: Split and full-body routines produced similar outcomes when training volume was matched.

Application and limit: Default to full-body sessions for two or three available days; allow splits for preference and scheduling. Full-body is a convenience choice, not a universal physiological winner.

Access: PubMed abstract. Source

[R20] Kassiano et al. (2022)

Does Varying Resistance Exercises Promote Superior Muscle Hypertrophy and Strength Gains? A Systematic Review. Journal of Strength and Conditioning Research 36(6):1753-1762.

Design: Eight studies, 241 participants, all young men.

Finding: Planned variation may aid regional development; excessive random variation and redundant choices can impede useful training continuity.

Application and limit: Keep a small stable exercise menu across a training block. Change exercises for a reason, with a separate performance history for each variation.

Access: PubMed abstract. Source | DOI

[R21] Advanced-methods network meta-analysis (2026)

The Effects of Advanced Resistance Training Prescription Methods on Strength, Power, Hypertrophy, and Performance Adaptations in Healthy Adults: A Systematic Review and Bayesian Network Meta-analysis. Sports Medicine; published April 8, 2026.

Design: Review and network meta-analysis of traditional training and seven advanced methods.

Finding: Advanced methods did not consistently outperform traditional training for strength, power, or hypertrophy in untrained to moderately trained adults.

Application and limit: Treat techniques such as advanced set structures as optional tools; simple progression remains a sound default. Highly trained populations were less well represented.

Access: Publisher abstract. Source | DOI

[R22] Bull et al. / WHO (2020)

World Health Organization 2020 guidelines on physical activity and sedentary behaviour. British Journal of Sports Medicine.

Design: International public-health guideline, with official WHO current guidance cross-check.

Finding: Adults should accumulate 150-300 minutes of moderate or 75-150 minutes of vigorous aerobic activity weekly, or an equivalent mix, alongside strengthening at least twice weekly.

Application and limit: Count walking outside formal workouts. Short strength sessions do not automatically fulfill aerobic targets; activity below the target still has value.

Access: Indexed guideline abstract and WHO official guidance. Source

[R23] Stefanovic, Werndle and Reljic (2026)

Effects of High-Intensity Interval Training on Cardiorespiratory Fitness and Cardiometabolic Health in Real-World Settings: A Systematic Review and Meta-Analysis. Sports 14(7):311; July 22, 2026.

Design: 31 studies, 1,119 participants; controlled and uncontrolled reports; search through December 2025.

Finding: Real-world HIIT was associated with improved cardiorespiratory fitness and selected cardiometabolic outcomes.

Application and limit: Offer intervals on suitable modes, including bikes. Within-group pooled changes do not establish superiority over continuous training or identify a universally best interval protocol.

Access: Indexed PubMed abstract. Source | DOI

[R24] Resistance-training variables review (2026)

Effects of resistance training variables on muscle hypertrophy in resistance-trained individuals: A systematic review. Journal of Bodywork and Movement Therapies 48:1149-1152; online August 21, 2026.

Design: Six studies in a qualitative synthesis; search through April 2025.

Finding: The review emphasizes training volume and effort as central contributors, with less consistent independent effects of other variables.

Application and limit: Recent corroboration, given limited weight because the synthesis is small. Do not optimize raw tonnage across exercises: load times reps is not a comparable hypertrophy dose across movements.

Access: Indexed publisher abstract. Source | DOI

Official guidance cross-checks

ACSM 2026 guideline announcement, WHO physical activity guidance, and CDC intensity guidance were checked alongside the scientific records. Recommendations about catalog priority, software fields, time budgeting and substitution behavior are the review's product-design inferences.